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St. Joseph's College of Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION-APRIL 2021

BC303S-ENZYMES

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL the questions:

I choose the best answer: (10X1=10)

1. Which class of enzymes do not exhibit stereospecificity

   a) Isomerases b) lyases c) hydrolases d) transferases

2. Maximum enzyme activity is observed at

   a) Acidic pH b) Neutral pH c) Basic pH d) Optimum pH

3. The group transferring coenzyme is

   a) DPN b) TPN c) FAD d) CoA

4. Lock and key mechanisms was proposed by

   a) Michaelis b) Koshland c) Fisher d) Menter

5. Protein moity of an enzyme is known as

   a) holoenzymes b) apoenzymes c) coenzymes d) enzyme

6. The isoenzymes of LDH

   a) Differ only in a single amino acid

   b) Differ in catalytic activity

   c) Exist in 5 forms depending on M and H monomer content

   d) Occur as monomers

7. Factors affecting enzyme activity

   a) Concentration b) pH c) Temperature d) All of the above

8. Line-weaver- Burk double reciprocal plot is related to

   a) Substrate concentration b) Enzyme activity

   c) Temperature d) Both A and B

9. An example of enzyme inhibition

   a) Reversible inhibition b) Irreversible inhibition

   c) Allosteric inhibition d) All of the above

10. Example of enzyme specificity

    a) Stereo specificity b) Reaction specificity

    c) Substrate specificity d) All of the above

II Say true or false: (5X1=5)

11. Ligases catalyst synthesis reaction by joining two molecules coupled with breakdown of pyrophosphate bond of ATP.
12. Koshland proposed lock & key model.
13. V_0 refers to initial velocity.
14. Trypsin is a proteolytic enzyme.
15. Mg is not a cofactor.

III Answer all the questions: (10X1=10)

16. What is inhibitor?
17. What is Vmax?
18. What do you mean by units of enzyme activity?
19. What do you mean by competitive inhibition?
20. What are isoenzymes?
21. What is Allosterism?
22. Give the enzymes of biological oxidation.
23. Give the significance of E.C. Numbers.

24. Feedback inhibition.
25. Define cofactors.

SECTION-B (4X5=20)

Answer any FOUR questions:

26. Write briefly on induced fit model of enzyme action.
27. Write briefly on specificity of enzymes
28. What are isoenzymes? Give examples.
29. Explain uncompetitive inhibition with examples.
30. Explain the Km value and its significance.
31. Write an account on immobilized enzymes.

SECTION-C (3X10=30)

Answer any THREE questions:

32. Explain the factors that influence enzyme activity.
33. Elaborate on enzyme inhibition and its types.
34. Explain the IUB classification of enzymes.
35. Write a note on the following
    (a) Acid base catalyst
    (b) Covalent catalyst
    (c) Metal ion catalyst
    (d) industrial application of enzymes.
36. Explain the various methods of enzymes purification.

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